

Definition:

A relation $\sim$ on a set S is called an *equivalence relation* if the following axioms hold:

1. (Reflexivity) $a \sim a$
2. (Symmetry) $a \sim b$ implies $b \sim a$
3. (Transitivity) $a \sim b$ and $b \sim c$ implies $a \sim c$

1. One of the most important equivalence relations we'll deal with this semester is $\sim_n$, defined by $a \sim_n b$ if $n \mid (b - a)$. For this equivalence relation...

(a) Re-phrase the reflexivity statement into a statement about a and n , then prove it.

(b) Re-phrase the symmetry statement into a statement about a , b , and n , then prove it.

(c) Re-phrase the transitivity statement into a statement about a , b , c and n , then prove it.

Consider the following relations. Either prove that the given relation is, in fact, an equivalence relation on the given set, or decide which of the three axioms fail to hold. If it is an equivalence relation, give a complete (non-repeating) list of the equivalence classes.

2. For $x, y \in \mathbb{R}$, we define $x \sim_d y$ if and only if $|x - y| < 5$.

3. For $x, y \in \mathbb{R}$, we define $x \sim_f y$ if and only if $\lfloor x \rfloor - \lfloor y \rfloor = 0$.

4. For $(a, b) \in \mathbb{R}^2$ and $(c, d) \in \mathbb{R}^2$, we define $(a, b) \sim_o (c, d)$ if and only if $a^2 + b^2 = c^2 + d^2$.